Exact Bounded Correlation

One-Phase Euler Coordinates for Stochastic Correlation, Derivative Pricing, and Joint Barriers

Complex–Itô Random-Walk Research Project

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We construct a stochastic-correlation framework in which the correlation state is the sine of the one-phase Euler phase of a scalar Gaussian factor: $\rho_t = X_t/\sqrt{X_t^2 + c^2}$ with $X_t = x_0 + \mu t + \sigma W_t$. Because the map is a monotone conjugacy, the framework inherits exact laws that no tractable competitor possesses jointly: a closed-form finite-time transition density; reflection-principle first-passage laws for correlation barriers; and exact prices for correlation derivatives — digitals, one-touches paid at expiry and at hit (a new closed form via a completing-the-square identity), and discretely monitored range accruals. For two assets driven by Brownian motions correlated through ρ_t, with the correlation driver independent of the asset drivers, the Dambis–Dubins–Schwarz theorem reduces every linear combination of the assets — spreads, baskets — to Brownian motion with a deterministic variance clock, collapsing joint barrier pricing to a scalar functional of the correlation path. The full law of that clock is computed deterministically by a Feynman–Kac resolvent and a Fourier series (the clock's support is bounded), verified against exact-skeleton Monte Carlo. The same resolvent prices continuous corridor occupation derivatives after an explicit atom separation, reproducing Lévy's arcsine law as a parameter-free anchor. Replacing the Brownian driver by an Ornstein–Uhlenbeck driver yields a mean-reverting correlation state with exact stationary density, and we prove the stationarity–tractability classification: static laws and the clock reduction survive, while closed-form first-passage laws are surrendered to the resolvent. All claims are verified by unit tests and reproducible notebooks; economic costs (non-stationarity under the Brownian driver, polynomial boundary tails, exclusion of leverage) are stated explicitly.

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1 Introduction

Correlation is the canonical bounded state variable of multi-asset finance, and it is empirically stochastic (Teng et al. 2016; Driessen et al. 2009). The modeling question is thirty years old and the answers split into two families, both with structural defects for pricing. Bounded *intrinsic* diffusions — the Jacobi process of van Emmerich (Emmerich 2006; Ackerer et al. 2018) — are mean-reverting but have no closed-form transition density, so likelihoods and prices need spectral expansions. *Transform* models — correlation as a bounded function of a Gaussian factor, as in the hyperbolic-tangent model of Teng, Ehrhardt, and Günther (Teng et al. 2016) — have exact marginals by change of variables, but the transform in use has never been given the path-level laws that exotic pricing consumes: barrier and one-touch structures under stochastic correlation are handled by approximation (Higgins 2014), and the first-passage problems of the underlying drivers are unsolved in closed form outside the Brownian case (Ricciardi and Sato 1988; Alili et al. 2005).

This paper closes that gap for one particular transform — the one whose path-level laws *are* closed form. The correlation state is

$$\rho_t = \sin\theta_t = \frac{X_t}{\sqrt{X_t^2 + c^2}}, \qquad \theta_t = \arctan(X_t/c), \qquad X_t = x_0 + \mu t + \sigma W_t,$$

the sine of the *one-phase Euler phase* of the companion paper (Complex–Itô Random-Walk Research Project 2026), where θ_t was proved to be the unique lossless phase coordinate of an affine complex Brownian process. The map $x \mapsto x/\sqrt{x^2 + c^2}$ is a smooth monotone conjugacy between the real line and $(-1, 1)$; every path property of X therefore transfers to ρ, and the Brownian driver's reflection principle becomes exact first-passage law for correlation barriers. No Jacobi, tanh-transformed-OU, or circular model has this, because their drivers' first-passage problems admit no reflection principle.

The contributions, each verified in the companion notebook (Complex–Itô Random-Walk Research Project 2026, companion notebook therein) and the reproducibility material of this paper, are:

1. **Exact kernels** (Section 2, Section 3): the finite-time transition density, the first-passage CDF and density, the perpetual hitting probability, and the two-sided band-exit law of ρ_t, all in terms of the Gaussian CDF; plus invariance of the entire class under Girsanov measure changes with constant correlation risk premium.
2. **Exact correlation derivatives** (Section 4): digital, one-touch paid at expiry, one-touch paid at hit — the last a new closed form obtained by completing the square in the inverse-Gaussian density — and discretely monitored range accruals.
3. **The variance-clock reduction** (Section 5): for two Bachelier assets correlated through ρ_t, any linear combination is, conditionally on the correlation path, Brownian motion with a deterministic clock. Joint barrier pricing collapses from three drivers to one scalar functional, with a sign theorem: correlation risk misprices spread and basket barriers in *opposite* directions.

4. **The full clock law** (Section 6): the clock's bounded support turns its Feynman–Kac characteristic function into a Fourier series, giving deterministic prices with two monitored discretization errors, statistically indistinguishable from exact simulation ($|z| < 1$ in all reported configurations).
5. **Corridor occupation** (Section 7): the same resolvent with an indicator potential, after explicit separation of the never-exit atom, prices continuous corridor digitals and calls; the driftless half-line case reproduces Lévy's arcsine law to 5×10^{-4}.
6. **The OU driver and the classification** (Section 8): an Ornstein–Uhlenbeck driver makes the correlation state mean-reverting with exact stationary density; we classify precisely what is kept (static laws, clock reduction, resolvent machinery) and what is surrendered (reflection-principle hitting laws), and verify the $\kappa \to 0^+$ degeneration into the Brownian results.

Section 9 states the novelty boundary: we do not claim the transform itself (bounded functions of Gaussian factors are established (Emmerich 2006; Teng et al. 2016)), nor a new correlation model in the empirical sense; the claim is the exact path-level and pricing content of this coordinate, which the audited literature does not contain, and the classification theorem of Section 8. The economic costs — non-stationarity under the Brownian driver, polynomial approach to the boundaries ± 1 (slower than the tanh model's exponential, the very objection raised in (Teng et al. 2016) against arctan-family maps), and the exclusion of leverage — are stated where they apply, not hidden.

1.1 Setup and conventions

Throughout, W is standard Brownian motion on a filtered probability space, Φ is the standard normal CDF, $c > 0$ is the chart scale, and

$$f(x) = \frac{x}{\sqrt{x^2 + c^2}}, \qquad f^{-1}(\rho) = \frac{c\,\rho}{\sqrt{1 - \rho^2}}, \qquad \frac{df^{-1}}{d\rho} = \frac{c}{(1 - \rho^2)^{3/2}}.$$

Quadratic variation is written $d[W]_t = dt$; no finite-increment identity is implied. Prices are at time 0 with constant rate $r \geq 0$ under a chosen pricing measure; Section 3 shows the exact class is invariant under the measure changes pricing requires.

2 The bounded correlation coordinate

Definition 2.1 (Bounded correlation coordinate)**.** Let $X_t = x_0 + \mu t + \sigma W_t$ with $\sigma > 0$, $c > 0$. The *bounded correlation coordinate* is $\rho_t = f(X_t)$, and $\theta_t = \arctan(X_t/c)$ is its *phase*, so $\rho_t = \sin\theta_t \in (-1, 1)$.

The coordinate is not an ad hoc choice: in the companion paper's classification (Complex–Itô Random-Walk Research Project 2026), θ_t is the *unique* time-independent lossless phase of an affine complex Brownian process $1 + iX_t/c = \sec\theta_t\, e^{i\theta_t}$, and $r(\theta) = \sec\theta$ is the polar equation of its supporting line. What the present paper adds is the financial reading of $\sin\theta$ as a correlation state and the pricing content that reading enables.

Proposition 2.1 (The autonomous correlation SDE)**.**

$$d\rho_t = \Big[\frac{\mu}{c}(1 - \rho_t^2)^{3/2} - \frac{3\sigma^2}{2c^2}\rho_t(1 - \rho_t^2)^2\Big]dt + \frac{\sigma}{c}(1 - \rho_t^2)^{3/2}\, dW_t.$$

Proof. Itô's formula with $f'(x) = c^2(x^2 + c^2)^{-3/2}$, $f''(x) = -3c^2 x(x^2 + c^2)^{-5/2}$, and $x^2 + c^2 = c^2/(1 - \rho^2)$. □

The diffusion coefficient vanishes like $(1-\rho^2)^{3/2}$ at the boundaries — faster than the Jacobi $\sqrt{1-\rho^2}$ — so ± 1 are unattainable in finite time (they correspond to $|X_t| = \infty$), and the coordinate respects correlation's natural state space without parameter restrictions, unlike the Jacobi family whose Feller condition constrains parameters (Emmerich 2006; Teng et al. 2016). The cost: approach to the boundaries is polynomial, so the model places more mass near strong correlation regimes than exponential-transform models; see Section 9.

Theorem 2.1 (Exact transition density)**.** *For $t > 0$ and $\rho \in (-1, 1)$,*

$$p_t(\rho \mid \rho_0) = \frac{c}{(1-\rho^2)^{3/2}} \, \frac{1}{\sigma\sqrt{2\pi t}} \exp\Big[-\frac{\big(f^{-1}(\rho) - f^{-1}(\rho_0) - \mu t\big)^2}{2\sigma^2 t}\Big].$$

Proof. Monotone change of variables of the Gaussian law of X_t. □

This is an exact *finite-time* law in a single closed form. For contrast, the closed-form density of the tanh-transformed modified-OU model (Teng et al. 2016) is its *stationary* density; its finite-time transition density is not closed form because its driver has none, and the Jacobi transition density requires spectral expansions (Ackerer et al. 2018). Figure 1 (left) verifies Theorem 2.1 against exact-skeleton simulation.

3 First-passage laws under the Brownian driver

Write $x(\rho) = f^{-1}(\rho)$ and, for an upper level $\rho^* > \rho_0$, $b = x(\rho^*) - x(\rho_0) > 0$ for the *conjugate barrier distance.*

Theorem 3.1 (Exact correlation first-passage law)**.** *Let $\tau_\rho = \inf\{t \geq 0 : \rho_t \geq \rho^*\}$. Then $\tau_\rho = \inf\{t \geq 0 : X_t \geq x(\rho^*)\}$ almost surely, and:*

(a) *for $T > 0$,*

$$\mathbb{P}(\tau_\rho \leq T) = \Phi\Big(\frac{\mu T - b}{\sigma\sqrt{T}}\Big) + e^{2\mu b/\sigma^2}\, \Phi\Big(-\frac{\mu T + b}{\sigma\sqrt{T}}\Big);$$

(b) *the (possibly defective) hitting-time density is inverse Gaussian,*

$$g_\tau(t) = \frac{b}{\sigma\sqrt{2\pi t^3}} \exp\Big[-\frac{(b-\mu t)^2}{2\sigma^2 t}\Big], \qquad \int_0^\infty g_\tau = \min(1, e^{2\mu b/\sigma^2});$$

(c) *the perpetual hitting probability is $\mathbb{P}(\tau_\rho < \infty) = 1$ for $\mu \geq 0$ and $e^{2\mu b/\sigma^2}$ for $\mu < 0$;* ***(d)*** *conditional moments for $\mu > 0$ are $\mathbb{E}[\tau \mid \tau < \infty] = b/\mu$ and $\mathrm{Var}(\tau \mid \tau < \infty) = b\sigma^2/\mu^3$.*

Proof. f is strictly increasing, so the events coincide; (a)–(d) are the reflection-principle facts of drifted Brownian motion (Borodin and Salminen 2002, sec. 1.1). □

The two-sided analogue follows from the scale function $s(x) = e^{-2\mu x/\sigma^2}$ ($s(x) = x$ for $\mu = 0$): for $\rho_\ell < \rho_0 < \rho_h$, $\mathbb{P}(\text{exit at } \rho_h) = (s(x_0) - s(x(\rho_\ell)))/(s(x(\rho_h)) - s(x(\rho_\ell)))$. Figure 1 (right) verifies (a) against bridge-corrected simulation.

Proposition 3.1 (Measure-change invariance)**.** *Under any equivalent martingale measure with constant market price of correlation risk λ, every law of this section and every price of Section 4 holds verbatim with μ replaced by $\mu^{\mathbb{Q}} = \mu + \sigma\lambda$.*

Proof. Girsanov with constant λ maps drifted Brownian motion to drifted Brownian motion (Girsanov 1960); the reflection principle and the conjugate map see only the drift. □

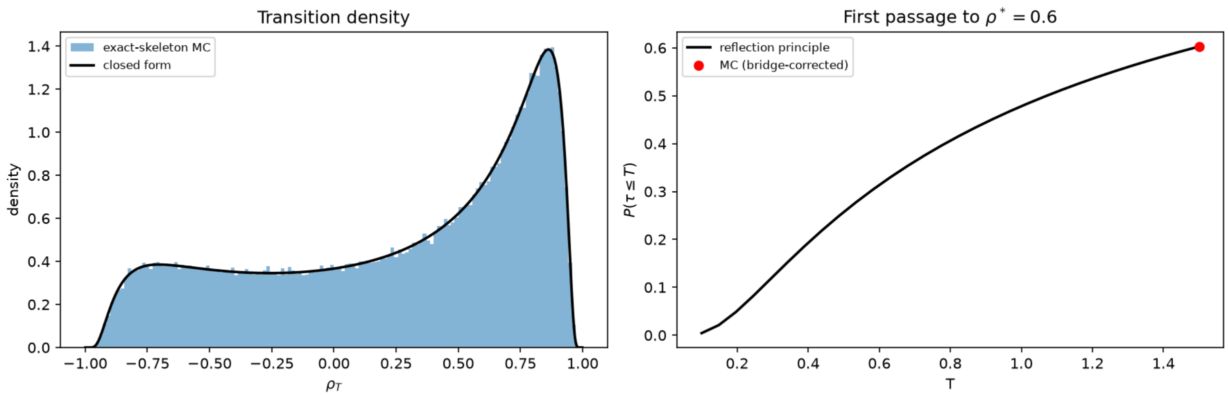

Figure 1: Exact kernels of the bounded correlation coordinate: transition density (left) and first-passage CDF (right), closed forms against exact-skeleton Monte Carlo. Seed 20260727.

Since correlation is not a traded asset, the pricing measure is a choice parameterized by λ; Proposition 3.1 says the exact-kernel class is *closed* under that choice. The correlation risk premium (Driessen et al. 2009) enters as one scalar drift parameter, and price sensitivities to it are closed-form comparative statics (no re-simulation).

4 Correlation derivatives

All four products below are priced in terms of Φ — no series, no transform inversion, no simulation. To our knowledge after the audit of Section 9, these are the first closed-form prices for correlation-barrier and correlation-range payoffs: the tractable competitors cover European-type payoffs only, and barrier/touch structures under stochastic correlation have approximations (Higgins 2014).

Proposition 4.1 (Digital). *$V_{\text{dig}} = e^{-rT}\Phi((\mu T - b)/(\sigma\sqrt{T}))$, paying $\mathbf{1}_{\{\rho_T \ge \rho^*\}}$ (any $\rho^* \in (-1,1)$).*

Proposition 4.2 (One-touch paid at expiry). *$V^T_{\text{touch}} = e^{-rT}\,\mathbb{P}(\tau_\rho \le T)$ with Theorem 3.1 (a), paying $\mathbf{1}_{\{\tau_\rho \le T\}}$ at T.*

Theorem 4.1 (One-touch paid at hit — a new closed form). *With $\gamma = \sqrt{\mu^2 + 2r\sigma^2}$,*

$$V^{\text{hit}}_{\text{touch}} = e^{-b(\gamma-\mu)/\sigma^2}\Big[\Phi\Big(\frac{\gamma T - b}{\sigma\sqrt{T}}\Big) + e^{2\gamma b/\sigma^2}\Phi\Big(-\frac{\gamma T + b}{\sigma\sqrt{T}}\Big)\Big],$$

paying 1 at τ_ρ if $\tau_\rho \le T$.

Proof. Completing the square, $(b-\mu t)^2/(2\sigma^2 t) + rt = (b-\gamma t)^2/(2\sigma^2 t) + b(\gamma-\mu)/\sigma^2$, so $e^{-rt}g^{(\mu)}_\tau(t) = e^{-b(\gamma-\mu)/\sigma^2}g^{(\gamma)}_\tau(t)$ and the right side integrates as Theorem 3.1 (a) with drift γ. □

Consistency limits, enforced as unit tests: $r \to 0^+$ recovers Proposition 4.2 undiscounted (both drift signs); $T \to \infty$ recovers the classical Laplace transform $e^{-b(\gamma-\mu)/\sigma^2}$; and $V^{\text{hit}}_{\text{touch}} \ge V^T_{\text{touch}} \ge V_{\text{dig}}$.

Proposition 4.3 (Discretely monitored range accrual). *For monitoring dates $0 < t_1 < \cdots < t_n = T$ and band (ρ_ℓ, ρ_h),*

$$V_{\text{range}} = \sum_{i=1}^{n} e^{-rt_i}\Big[\Phi\Big(\frac{x(\rho_h) - x_0 - \mu t_i}{\sigma\sqrt{t_i}}\Big) - \Phi\Big(\frac{x(\rho_\ell) - x_0 - \mu t_i}{\sigma\sqrt{t_i}}\Big)\Big],$$

paying $\sum_i \mathbf{1}_{\{\rho_\ell \le \rho_{t_i} \le \rho_h\}}$. Either level may be ∓ 1, recovering one-sided digitals.

Daily monitoring is how range accruals are written, so Proposition 4.3 is the practical product; the *continuous* corridor is the harder object and is solved in Section 7.

The paper notebook reports all prices against bridge-corrected Monte Carlo (within 2 standard errors), and plots the one-touch price against the correlation risk premium λ of Proposition 3.1 — a closed-form comparative static that the approximation literature cannot produce.

5 Joint two-asset barriers: the variance-clock reduction

5.1 Model

Two traded forward prices (martingales under the forward measure, so no drifts appear), correlated through the bounded coordinate:

$$dF_1 = \sigma_1\, dW_1, \qquad dF_2 = \sigma_2(\rho_t\, dW_1 + \sqrt{1-\rho_t^2}\, dW_2),$$

with $\rho_t = f(X_t)$ as before. **Structural assumption:** the correlation driver W is independent of the asset drivers (W_1, W_2) — the van Emmerich/Teng construction (Emmerich 2006; Teng et al. 2016). The cost is the exclusion of leverage (correlation does not respond to asset moves); see Section 9.

Lemma 5.1 (Conditional Gaussianity)**.** *Conditional on the correlation path $\rho_{[0,T]}$, the pair (F_1, F_2) is Gaussian with deterministic covariance determined by $\rho_{[0,T]}$.*

Proof. Itô integrals with (conditionally) deterministic integrands are Gaussian; independence of W makes conditioning free. □

Theorem 5.1 (Deterministic variance clock)**.** *Let $Z = \alpha_1 F_1 + \alpha_2 F_2$ (spread $\alpha = (1,-1)$, basket $\alpha = (1,1)$), $z_0 = \alpha_1 F_{1,0} + \alpha_2 F_{2,0}$. Conditionally on $\rho_{[0,T]}$,*

$$Z_t \overset{\text{law}}{=} z_0 + B_{v(t)}, \qquad v(t) = \int_0^t q(\rho_s)\, ds,$$

$$q(\rho) = \alpha_1^2\sigma_1^2 + \alpha_2^2\sigma_2^2 + 2\alpha_1\alpha_2\sigma_1\sigma_2\, \rho.$$

Proof. Z is a continuous local martingale with quadratic variation $\int q(\rho_s)\, ds$, deterministic under conditioning; the Dambis–Dubins–Schwarz theorem (Karatzas and Shreve 1991, Thm. 3.4.6) applies. □

Corollary 5.1 (Scalar reduction of joint barrier laws)**.** *For an upper barrier $d > z_0$,*

$$\mathbb{P}\Big(\sup_{t\le T} Z_t \ge d \;\Big|\; \rho_{[0,T]}\Big) = 2\Big(1 - \Phi\Big(\frac{d - z_0}{\sqrt{v(T;\rho)}}\Big)\Big),$$

and analogously for lower barriers and the joint law of $(\sup Z, Z_T)$. Every claim measurable in the running extrema of Z prices as $e^{-rT}\mathbb{E}_\rho[g(v(T;\rho))]$ with g closed form.

The three-driver pricing problem collapses to the law of one scalar functional. The notebook verifies the theorem itself, not just its consequences: a full three-driver simulation with bridge correction matches the scalar reduction (z-scores -1.16, $+1.19$ at two step sizes), with the reduction's standard error six times smaller — an effective-dimension reduction, not a relabeling.

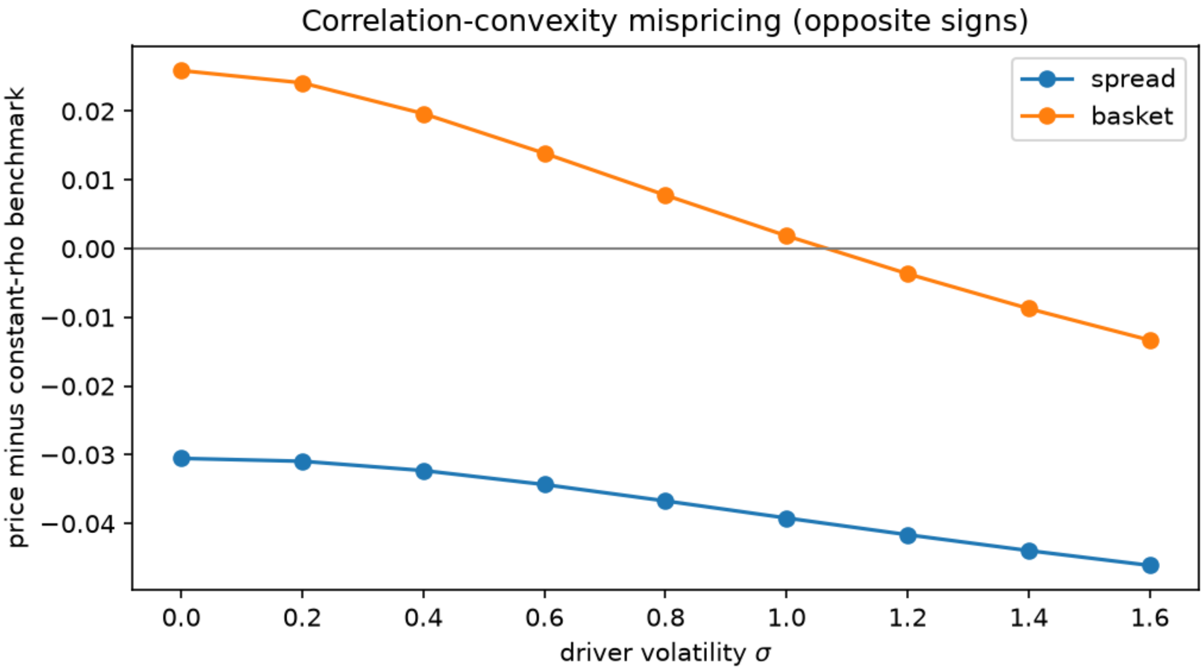

Figure 2: Correlation-convexity mispricing: stochastic-correlation one-touch price minus the constant-correlation benchmark, for spread and basket barriers. Opposite signs, as Corollary 5.2 requires. Seed 20260727.

Corollary 5.2 (Sign theorem: spread vs. basket)**.** *The q-coefficient of ρ is $2\alpha_1\alpha_2\sigma_1\sigma_2$: negative for the spread, positive for the basket. Correlation risk therefore moves spread-barrier and basket-barrier prices in opposite directions, and constant-correlation pricing misprices the two in opposite senses.*

Figure 2 exhibits Corollary 5.2: the gap between stochastic- and constant-correlation one-touch prices has opposite signs for spread and basket across the correlation-volatility range. Since $v \mapsto 2(1-\Phi(d/\sqrt{v}))$ changes convexity in v, the *sign* of the Jensen gap is parameter-dependent and is exhibited numerically rather than asserted as a theorem.

Working with forward prices is what makes Theorem 5.1 exact: with nonzero drifts the running supremum depends on the whole variance path, not only on $v(T)$.

6 The full clock law by Feynman–Kac

Corollary 5.1 prices everything through the law of $v(T;\rho) = \beta T + \alpha A_T$ with $\beta = \alpha_1^2\sigma_1^2 + \alpha_2^2\sigma_2^2$, $\alpha = 2\alpha_1\alpha_2\sigma_1\sigma_2$ and $A_T = \int_0^T \rho_s\, ds$. The moments of A_T are deterministic quadratures (Gauss–Hermite over the Gaussian factor, Gauss–Legendre over the time simplex — the naive square rule has a diagonal ridge; the triangle-aware rule converges to machine precision). For the full law:

Theorem 6.1 (Bounded support Fourier density)**.** *Let $u(t,x) = \mathbb{E}_x[e^{i\xi\int_0^t f(X_s)\,ds}]$. Then*

$$\partial_t u = \tfrac{\sigma^2}{2} u_{xx} + \mu u_x + i\xi f(x)\, u, \qquad u(0,x) = 1,$$

and $\phi(\xi) = u(T, x_0)$ *is the characteristic function of* A_T*. Since* $|f| < 1$*, the support of* A_T *is bounded in* $(-T, T)$*, and on* $[-T, T]$ *the density is*

$$p(a) = \frac{1}{2T} \sum_{k=-\infty}^{\infty} \phi\Big(\frac{k\pi}{T}\Big) e^{-ik\pi a/T}.$$

Proof. Feynman–Kac (Kac 1949) applies since the potential is bounded; the Fourier representation is the standard series on a bounded interval. □

One PDE solve per Fourier mode (Krylov time integration on a truncated domain; the Dirichlet localization error is checked by doubling the margin, cf. (Cont and Voltchkova 2005)). The price is

$$V_{\mathrm{FK}} = e^{-rT} \int_{-T}^{T} g(\beta T + \alpha a)\, p_K(a)\, da,$$

with two monitored discretization errors (K, grid). In all reported configurations the deterministic price is statistically indistinguishable from the exact-simulation benchmark ($|z| < 1$); a Gamma moment match on the quadrature moments remains as a fast screening tool whose error against the resolvent price is measured per parameter set (-5×10^{-5} to -2×10^{-3} across the correlation-volatility range).

7 Corridor occupation

The continuous corridor pays on the occupation time $\mathcal{O}_T = \int_0^T \mathbf{1}_{\{x_\ell \le X_t \le x_h\}}\, dt$ — exactly the occupation of the correlation band $\{\rho_\ell \le \rho_t \le \rho_h\}$. The plain corridor bond is linear in $\mathcal{O}_T$ and prices by quadrature of Gaussian probabilities (an anchor, not a contribution); anything nonlinear — a corridor digital $\mathbf{1}_{\{\mathcal{O}_T \ge \kappa T\}}$ or call $(\mathcal{O}_T/T - \kappa)^+$ — needs the law.

The resolvent of Theorem 6.1 applies with potential $i\xi \mathbf{1}_{[x_\ell, x_h]}$. Two structural facts drive the computation:

Proposition 7.1 (The never-exit atom)**.** *A bounded corridor always has an atom at* $\mathcal{O}_T = T$ *(never exit), computable deterministically by the absorbing-boundary PDE on the corridor; an unbounded corridor with drift has one in closed form from Theorem 3.1 (c). At the Fourier nodes* $\xi_k = 2\pi k/T$ *the atom contributes its mass as a constant* ($e^{i\xi_k T} = 1$)*, so it is subtracted from every mode and the Fourier series represents only the continuous part.*

Theorem 7.1 (Arcsine anchor)**.** *For* $\mu = 0$*, corridor* $[0, \infty)$*,* $x_0 = 0$*, the occupation fraction follows Lévy's arcsine law* $\mathbb{P}(\mathcal{O}_T \le sT) = \frac{2}{\pi} \arcsin \sqrt{s}$ *(Lévy 1939). The resolvent pipeline reproduces it: tail probabilities to* 5×10^{-4} *(Figure 3).*

The notebook prices corridor digitals and calls against exact-skeleton simulation; the residual gap ($\sim 7 \times 10^{-3}$) is fully explained by two identified opposite-sign biases (simulation overstates occupation through bridge misses; Fourier truncation biases slightly downward), and is recorded as such in the reproducibility material — not reported as statistical agreement.

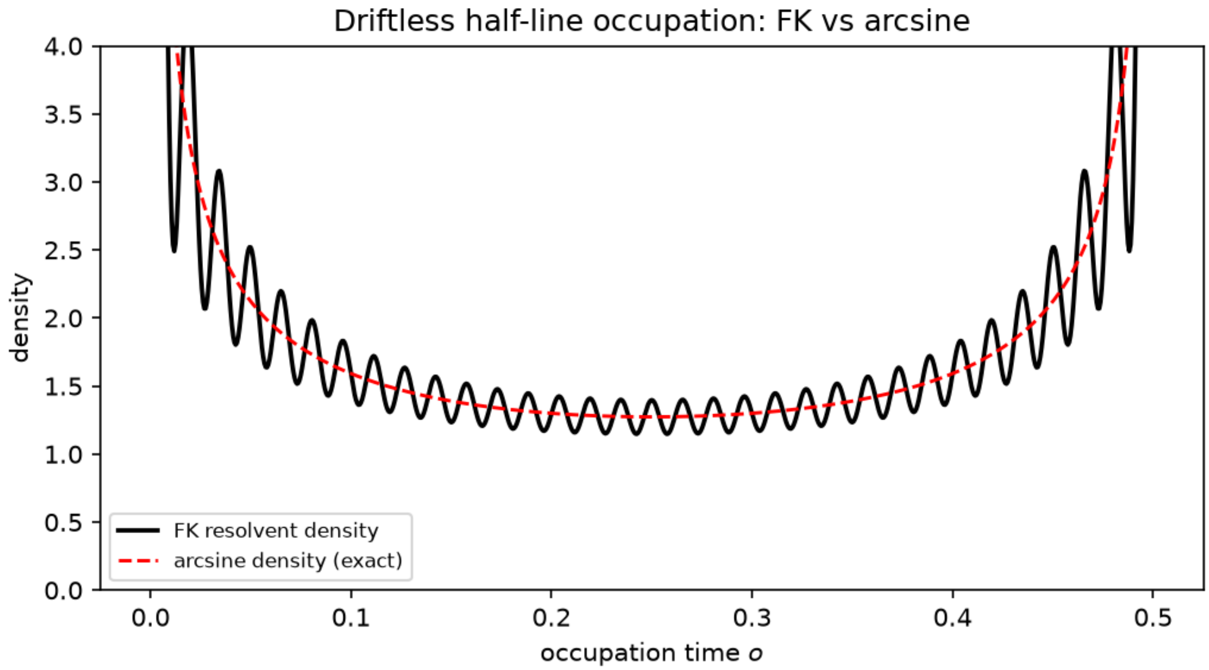

Figure 3: Driftless half-line occupation: the FK resolvent density against Lévy's exact arcsine density — a parameter-free, non-Gaussian benchmark. Seed 20260727.

8 The OU driver and the stationarity–tractability classification

Replace the driver by $dX_t = \kappa(\bar{x} - X_t)\,dt + \sigma\,dW_t$, $\kappa > 0$. The correlation coordinate inherits mean reversion and an exact stationary density (Gaussian stationary law pushed through f), answering the standard objection to non-stationary correlation models (Teng et al. 2016).

Theorem 8.1 (Stationarity–tractability classification)**.** *For $\rho_t = f(X_t)$ with X a scalar Gaussian factor:*

object	*BM driver*	*OU driver*
stationarity	*none*	*exact stationary density*
transition density	*closed form*	*closed form*
digital / discrete accrual	*closed form*	*closed form*
first-passage laws	*closed form (reflection)*	*no closed form; FK resolvent*
DDS variance clock	*exact*	*exact (driver-agnostic)*
clock moments / full law	*quadrature / FK*	*quadrature / FK (OU generator)*
measure change, constant λ	$\mu \to \mu + \sigma\lambda$	$\bar{x} \to \bar{x} + \sigma\lambda/\kappa$

Proof. Static rows: OU transitions are Gaussian, and the conjugate map is deterministic. First-passage row: OU hitting laws admit only series representations (Ricciardi and Sato 1988; Alili et al. 2005); hitting probabilities are path properties and the monotone map cannot repair what the generator lacks. Clock rows: Theorem 5.1 conditions on the realized path and never sees the generator. Measure-change row: Girsanov with constant λ shifts the OU drift by a constant, equivalently the long-run mean. □

Corollary 8.1 (The frontier)**.** *In this model class, reflection-principle hitting laws and stationarity are mutually exclusive: a Gaussian factor with a Gaussian stationary law is not Brownian, and only Brownian motion has reflection-principle hitting laws. Mean reversion is bought by surrendering closed-form barrier laws to the resolvent; everything static and everything clock-based survives.*

The $\kappa \to 0^+$ degeneration anchors the implementation: every OU formula reproduces its Brownian counterpart (density to 2×10^{-7}, occupation moments to 6×10^{-8}), and the OU spread one-touch price from the resolvent matches exact OU simulation (z = +0.92).

9 Novelty boundary and related work

Established, not claimed. Bounded correlation as a transform of a Gaussian factor: van Emmerich (Emmerich 2006); tanh of a modified OU with closed-form *stationary* density and quanto pricing by conditional Monte Carlo: Teng–Ehrhardt–Günther (Teng et al. 2016); Jacobi correlation and Jacobi volatility with spectral methods (Ackerer et al. 2018); circular correlation models with approximate likelihood (Majumdar 2024); barrier/touch *approximations* under stochastic (spot–vol) correlation (Higgins 2014); the arctangent map as a numerical compactification in finance (Duffie and Kan 1996; Hout and Snoeijer 2021); arctangent-transformed Brownian motion as an exact SDE example (Särkkä and Solin 2019); Bachelier pricing context (Schachermayer and Teichmann 2008; Choi et al. 2022; CME Group 2020); exact simulation beyond transform methods (Beskos and Roberts 2005). The one-phase Euler chart itself is the companion paper (Complex–Itô Random-Walk Research Project 2026).

What the audit found open, and what is claimed here. (i) Exact finite-time kernels and reflection-principle first-passage laws for a bounded correlation state (Theorem 2.1, Theorem 3.1) — the competitors have stationary-only densities, spectral densities, or approximate ones. (ii) Exact prices for correlation-barrier and correlation-range payoffs, including the pay-at-hit closed form of Theorem 4.1 (Section 4, Section 7) — previously approximation territory. (iii) The scalar variance-clock reduction with its sign theorem (Theorem 5.1, Corollary 5.2) and the deterministic full clock law (Theorem 6.1). (iv) The stationarity–tractability classification (Theorem 8.1). Teng et al. (Teng et al. 2016) considered and rejected an arctan-family correlation map for its slow polynomial approach to ± 1; we accept that cost explicitly (it is the price of the reflection principle), and note the rejection concerned a different member of the family ($(2/\pi)\arctan$ of the factor, not $\sin \circ \arctan$), and was an estimation argument, not a pricing one.

Costs and exclusions, stated. Under the Brownian driver the coordinate is non-stationary ($\rho_t \to \operatorname{sgn} \mu$ a.s. for $\mu \neq 0$; recurrent extremes for $\mu = 0$); the OU driver of Section 8 answers this at the cost of Corollary 8.1. Boundary approach is polynomial. The reduction of Section 5 excludes leverage ($W \perp (W_1, W_2)$). Correlation is not traded; prices are conditional on a chosen risk premium (Proposition 3.1). Estimation and calibration are deferred, though exact likelihoods are immediate from Theorem 2.1.

10 Conclusion

One bounded change of variables — the sine of the one-phase Euler phase — carries Brownian exactness through every level of the stochastic correlation problem: kernels, first passage, derivative prices, joint barrier laws, occupation laws, and, with the OU driver, a classified stationarity frontier. The framework's value is not a new transform but the closed-form and deterministic content the transform inherits from the reflection principle and the Feynman–Kac resolvent — content the tractable literature prices only by approximation.

Reproducibility

All numerical claims are produced by the companion notebook (`notebooks/phase6_correlation_paper.ipynb`, seed 20260727, runtime under one minute) from tested repository modules (`phase6_correlation.py`, `phase6_joint.py`, `phase6_fk.py`, `phase6_corridor.py`, `phase6_ou.py`; 77 unit tests). Key checks: transition density and first-passage CDF vs. exact-skeleton Monte Carlo (Figure 1); derivative ordering $V_{\text{dig}} \leq V^{T}_{\text{touch}} \leq V^{\text{hit}}_{\text{touch}}$; full three-driver vs. scalar reduction; FK price vs. benchmark ($|z| < 1$); arcsine tail to 5×10^{-4}; $\kappa \to 0^{+}$ anchors; OU price vs. benchmark ($z = -0.14$). Figures and machine-readable checks are in `figures/` and `tables/` (`verification_summary.csv`).

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